TINGSONG XIAO

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Homepage; Google Scholar

RESEARCH INTERESTS

AI for Health, Deep Learning, Large Language Models, Data Mining, Time Series, and Event Sequence Modeling.

EDUCATION

Doctor of Philosophy, Ph.D. | **GPA: 3.86/4.00** | ***Computer Science*** — Aug. 2022 – Now
University of Florida — Gainesville, FL, U.S.

Master of Science, M.S. | **GPA: 3.86/4.00** | ***Computer Science*** — Aug. 2022 – Dec. 2024
University of Florida (Conferred during Ph.D. program) — Gainesville, FL, U.S.

Bachelor of Engineering, B.E. | **GPA: 3.92/4.00** | ***Computer Science*** — Aug. 2017 – June 2021
University of Electronic Science and Technology of China (UESTC) — Chengdu, China

EXPERIENCE

Applied Researcher Intern — May 2025 – Aug. 2025
eBay Inc. — San Jose, CA, U.S.

- Developed an **efficient LLM-based pipeline** to extract structured key-value pairs (e.g., brand, model, color) from **unstructured product descriptions** provided by sellers.
- Built a scalable prompt-driven knowledge elicitation and **LLM distillation** framework that replaces heuristic rule-based filters with a compact student model for real-time inference.
- Conducted finetuning with **multi-strategy distillation** (response-based, logit-based, and hybrid) from powerful teacher models (e.g., GPT-4, LLaMA3-70B) into smaller LLMs.
- Delivered a student model that achieves **over 90% F1 score** on key-value extraction tasks across 10 product categories while reducing inference latency by over **5×**, enabling deployment **at scale** to millions of listings.

Research Assistant — Aug. 2022 – Dec. 2026
University of Florida — Gainesville, FL, U.S.

- Conducted collaborative research in **AI for Health**, leveraging data mining and predictive modeling to analyze longitudinal electronic health records **(time seires, event sequences)** for patient safety and disease progression modeling.
- Introduced XTSFormer, an **efficient and scalable Transformer-based** Temporal Point Process model designed for **irregular sequential events** analysis, incorporating multi-scale temporal interactions of event features.
- Developed two novel components: a feature-based cycle-aware time positional encoding, capturing complex temporal patterns by integrating feature and **cyclical information**, and a cross-temporal-scale attention mechanism, improving **time efficiency** over standard all-pair attention.
- Demonstrated superior performance with **5x faster training time** compared to vanilla Transformers on event sequences with **a length of 100,000** and achieving **3.3% higher accuracy and F1-score** on real-world event sequences.
- Published papers in top-tier conferences, including AAAI 2025 and NeurIPS 2024, and received the AAAI 2025 Scholarship and Volunteer Award as well as the ACM SIGSPATIAL 2023 Best Paper Award.

Research Assistant — Sep. 2021 – July 2022
University of Electronic Science and Technology of China — Chengdu, China

- Studied medical image classification and the **interpretability** of graph neural networks (GNNs), focusing on Alzheimer's disease (AD) diagnosis.
- Proposed and implemented a **dual-graph learning convolutional network** (dGLCN) within the GCN framework to address interpretability challenges by simultaneously learning subject and feature graphs.
- Achieved state-of-the-art performance on the ADNI dataset with a 3.2% improvement in accuracy over the best comparison method, and provided interpretability of **important nodes and features** during the graph node classification task.
- Published papers in leading journals and conferences, and received the MICCAI 2022 Student Travel Award.

Selected Publications

- **Tingsong Xiao**, Zelin Xu, Wenchong He, Zhengkun Xiao, et al. "XTSFormer: Cross-Temporal-Scale Transformer for Irregular-Time Event Prediction in Clinical Applications." *The 39th Annual AAAI Conference on Artificial Intelligence (AAAI 2025).* **AAAI Scholarship and Volunteer Award.**
- **Tingsong Xiao**, Lu Zeng, Xiaoshuang Shi, Xiaofeng Zhu, and Guorong Wu. "Dual-graph Learning Convolutional Networks for Interpretable Alzheimer's Disease Diagnosis." *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI 2022). Springer, Cham, 2022.* **Early Accepted; Oral Paper; MICCAI Student Travel Award.**
- Zelin Xu, Yupu Zhang, **Tingsong Xiao**, et al. "CoastalBench: A Decade-Long High-Resolution Dataset to Emulate Complex Coastal Processes." *The Forty-Second International Conference on Machine Learning (ICML 2025).*
- Zelin Xu, **Tingsong Xiao**, Wenchong He, Yu Wang, et al. "Spatial-Logic-Aware Weakly Supervised Learning for Flood Mapping on Earth Imagery." *The 38th Annual AAAI Conference on Artificial Intelligence (AAAI 2024).*
- Wenchong He, Zhe Jiang, and **Tingsong Xiao**, et al. "A Hierarchical Spatial Transformer for Large Numbers of Point Samples in Continuous Space." *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023)*.
- Zelin Xu, **Tingsong Xiao**, Wenchong He, Yu Wang, and Zhe Jiang. "Spatial Knowledge-Infused Hierarchical Learning: An Application in Flood Mapping on Earth Imagery." *ACM International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL 2023)*. **Best Paper Award.**
- Jing Hu, Xincheng Wang and Ziheng Liao, **Tingsong Xiao***. "L-GCN: Local Graph Neural Network for 3D Point Clouds Classification." *IEEE International Conference on Multimedia and Expo (ICME 2023).* **(* corresponding author)**.
- Lu Zeng, Hengxin Li, **Tingsong Xiao**, Fumin Shen and Zhi Zhong. "Graph convolutional network with sample and feature weights for Alzheimer's disease diagnosis." *Information Processing & Management (IP&M) 59.4 (2022): 102952.*
- Xiaochen Wang, **Tingsong Xiao**, Jie Tan, Deqiang Ouyang and Jie Shao. "MRMRP: multi-source review-based model for rating prediction." *International Conference on Database Systems for Advanced Applications (DASFAA). Springer, Cham, 2020.*
- Xiaochen Wang, **Tingsong Xiao** and Jie Shao. "EMRM: Enhanced Multi-source Review-Based Model for Rating Prediction." *International Conference on Knowledge Science, Engineering and Management (KSEM). Springer, Cham, 2021.*

Honors and Awards

AAAI Scholarship and Volunteer Award — Feb. 2025
To recognize outstanding contributions and active participation through academic excellence in the 39th AAAI Conference on Artificial Intelligence (AAAI-25).

SDM Doctoral Forum Award — Mar. 2024
A competitive NSF-funded venue recognizing promising data mining researchers for their intellectual merit and broader impacts.

ACM SIGSPATIAL Best Paper Award — Nov. 2023
Awarded annually to the best paper at the ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems.

Gartner Group Graduate Fellowship — Apr. 2023, Apr. 2025
Recognizes outstanding graduate students at the University of Florida for academic excellence and research potential.

MICCAI 2022 Student Travel Award Mar. 2022

To reward first author students of the highest-quality MICCAI papers.

Honorable Mention of Mathematical Contest In Modeling Feb. 2020

Top 21% of the contest administered by COMAP and sponsored by SIAM and INFORMS.

INVITED JOURNAL AND CONFERENCE REVIEWERS

- ACM Transactions on Intelligent Systems and Technology (TIST)
- IEEE Transactions on Image Processing (TIP)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Transactions on Computational Social Systems (TCSS)
- IEEE Transactions on Multimedia (TMM)
- Information Processing & Management (IP&M)
- Future Generation Computer Systems
- Information Sciences
- Neural Networks
- Neural Processing Letters
- Pattern Recognition Letters
- IEEE International Symposium on Biomedical Imaging (ISBI)
- IEEE International Conference on Data Mining (ICDM)
- International Conference on Machine Learning (ICML)
- International Conference on Learning Representations (ICLR)
- Conference on Neural Information Processing Systems (NeurIPS)
- Association for the Advancement of Artificial Intelligence (AAAI)
- International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)

SKILLS

- Programming Languages: Python, Matlab, C.
- Deep Learning Framework: PyTorch, TensorFlow, Keras.